

EDA

May 6, 2026

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[5]: # =====
# STEP 2 - EXPLORATORY DATA ANALYSIS (EDA)
# Project: Sales & Profit Analysis - Superstore Dataset
# Tool: Python (Matplotlib & Seaborn)
# =====

import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Settings for better looking charts
plt.rcParams['figure.figsize'] = (12, 6)
plt.rcParams['font.size'] = 12
sns.set_theme(style='whitegrid')

# Load the clean dataset
df = pd.read_csv('../Clean_Superstore.csv')

print(" Clean dataset loaded successfully!")
print(f"Shape: {df.shape}")
print(f"\nFirst 5 Rows:")
print(df.head())
print(f"\nBasic Statistics:")
print(df[['Sales', 'Profit', 'Quantity', 'Discount']].describe())
```

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Clean dataset loaded successfully!
Shape: (9733, 25)
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First 5 Rows:

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	\
0	5848	CA-2015-126725	2015-11-17	2015-11-21	Standard Class	BS-11665	
1	5694	US-2015-138093	2015-10-12	2015-12-16	Standard Class	KM-16225	
2	5971	CA-2014-141726	2014-07-20	2014-07-22	First Class	CC-12145	
3	8965	CA-2016-157280	2016-11-05	2016-11-07	First Class	LW-17125	
4	2019	CA-2015-113110	2015-03-19	2015-03-23	Standard Class	BK-11260	

	Customer Name	Segment	Country	City	...	\
0	Brian Stugart	Consumer	United States	San Diego	...	

1	Kalyca Meade	Corporate	United States	Baltimore	...
2	Charles Crestani	Consumer	United States	San Diego	...
3	Liz Willingham	Consumer	United States	Virginia Beach	...
4	Berenike Kampe	Consumer	United States	San Bernardino	...

	Sub-Category	Product Name	Sales	\
0	Furnishings	Advantus Panel Wall Certificate Holder - 8.5x11	36.600	
1	Paper	Astroparche Fine Business Paper	26.400	
2	Paper	Xerox 1897	19.920	
3	Furnishings	Tenex Chairmat w/ Average Lip, 45" x 53"	54.686	
4	Supplies	Acme Elite Stainless Steel Scissors	54.686	

	Quantity	Discount	Profit	Order Year	Order Month	Order Month Name	\
0	3.0	0.0	15.3720	2015	11	November	
1	5.0	0.0	12.6720	2015	10	October	
2	4.0	0.0	9.7608	2014	7	July	
3	5.0	0.0	75.6800	2016	11	November	
4	4.0	0.0	8.6736	2015	3	March	

	Order Quarter
0	4
1	4
2	3
3	4
4	1

[5 rows x 25 columns]

Basic Statistics:

	Sales	Profit	Quantity	Discount
count	9733.000000	9733.000000	9733.000000	9733.000000
mean	217.741002	28.149159	3.766156	0.157117
std	606.829770	213.527084	2.198853	0.204508
min	0.444000	-3839.990400	1.000000	0.000000
25%	18.312000	1.932000	2.000000	0.000000
50%	54.686000	8.671000	3.000000	0.200000
75%	194.352000	27.435200	5.000000	0.200000
max	22638.480000	8399.976000	14.000000	0.800000

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[6]: print("=" * 50)
print("CHART 1 - SALES & PROFIT BY CATEGORY")
print("=" * 50)

# Calculate total sales and profit per category
category_data = df.groupby('Category')[['Sales', 'Profit']].sum().reset_index()
print(category_data)
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# Create the chart
fig, axes = plt.subplots(1, 2, figsize=(14, 6))

# Sales by Category
axes[0].bar(category_data['Category'], category_data['Sales'],
            color=['#2196F3', '#4CAF50', '#FF9800'])
axes[0].set_title('Total Sales by Category')
axes[0].set_xlabel('Category')
axes[0].set_ylabel('Total Sales ($)')
for i, v in enumerate(category_data['Sales']):
    axes[0].text(i, v + 1000, f'${v:,.0f}', ha='center', fontweight='bold')

# Profit by Category
axes[1].bar(category_data['Category'], category_data['Profit'],
            color=['#2196F3', '#4CAF50', '#FF9800'])
axes[1].set_title('Total Profit by Category')
axes[1].set_xlabel('Category')
axes[1].set_ylabel('Total Profit ($)')
for i, v in enumerate(category_data['Profit']):
    axes[1].text(i, v + 500, f'${v:,.0f}', ha='center', fontweight='bold')

plt.suptitle('Sales & Profit by Category', fontsize=16, fontweight='bold')
plt.tight_layout()
plt.savefig('chart1_category.png', dpi=150, bbox_inches='tight')
plt.show()
print(" Chart 1 saved as 'chart1_category.png'")

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CHART 1 - SALES & PROFIT BY CATEGORY
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	Category	Sales	Profit
0	Furniture	679398.0927	15236.7182
1	Office Supplies	681205.0130	120088.4138
2	Technology	758670.0700	138650.6364

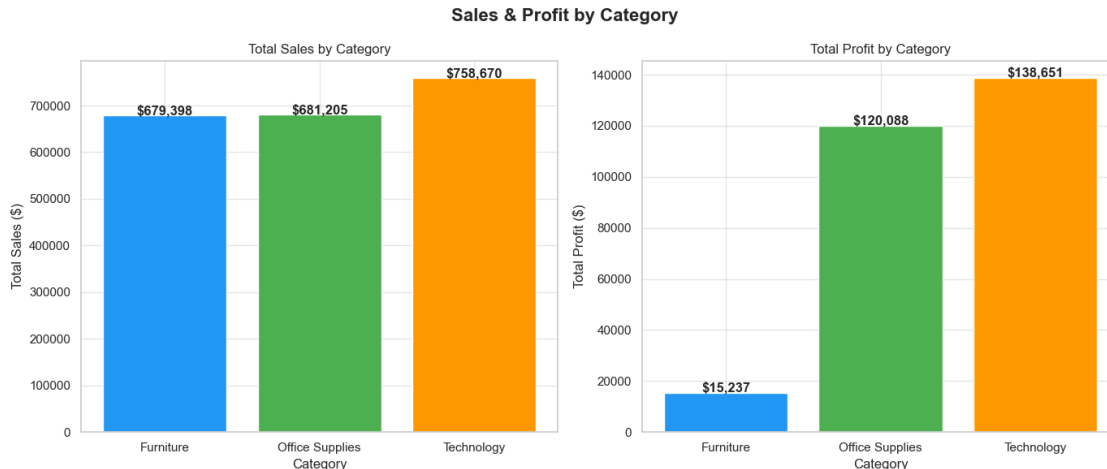


Chart 1 saved as 'chart1_category.png'

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[7]: print("=" * 50)
print("CHART 2 - SALES & PROFIT BY REGION")
print("=" * 50)

# Calculate total sales and profit per region
region_data = df.groupby('Region')[['Sales', 'Profit']].sum().reset_index()
print(region_data)

# Create the chart
fig, axes = plt.subplots(1, 2, figsize=(14, 6))

# Sales by Region
axes[0].bar(region_data['Region'], region_data['Sales'],
            color=['#E91E63', '#9C27B0', '#3F51B5', '#00BCD4'])
axes[0].set_title('Total Sales by Region')
axes[0].set_xlabel('Region')
axes[0].set_ylabel('Total Sales ($)')
for i, v in enumerate(region_data['Sales']):
    axes[0].text(i, v + 1000, f'${v:,.0f}', ha='center', fontweight='bold')

# Profit by Region
axes[1].bar(region_data['Region'], region_data['Profit'],
            color=['#E91E63', '#9C27B0', '#3F51B5', '#00BCD4'])
axes[1].set_title('Total Profit by Region')
axes[1].set_xlabel('Region')
axes[1].set_ylabel('Total Profit ($)')
for i, v in enumerate(region_data['Profit']):
    axes[1].text(i, v + 500, f'${v:,.0f}', ha='center', fontweight='bold')
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plt.suptitle('Sales & Profit by Region', fontsize=16, fontweight='bold')
plt.tight_layout()
plt.savefig('chart2_region.png', dpi=150, bbox_inches='tight')
plt.show()
print(" Chart 2 saved as 'chart2_region.png'")
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CHART 2 - SALES & PROFIT BY REGION

	Region	Sales	Profit
0	Central	452748.4252	40903.6300
1	East	620185.5330	94961.5656
2	South	348416.5805	40310.4860
3	West	697922.6370	97800.0868

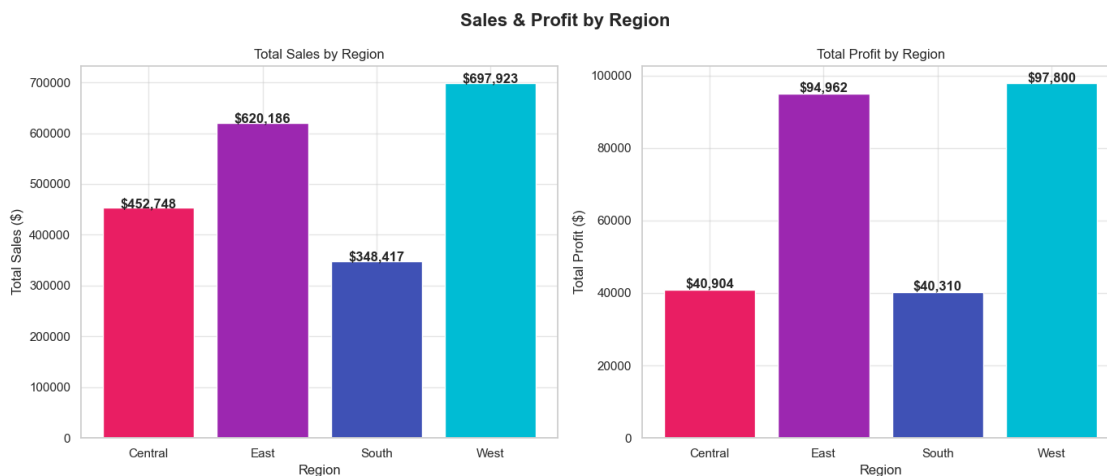


Chart 2 saved as 'chart2_region.png'

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[8]: print("=" * 50)
print("CHART 3 - MONTHLY SALES TREND")
print("=" * 50)

# Calculate total sales per month per year
monthly_sales = df.groupby(['Order Year', 'Order Month'])['Sales'].sum().
    ↪reset_index()
print(monthly_sales.head(12))

# Create the chart
fig, ax = plt.subplots(figsize=(16, 6))

colors = {2014: '#E91E63', 2015: '#2196F3', 2016: '#4CAF50', 2017: '#FF9800'}
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for year in sorted(df['Order Year'].unique()):
    year_data = monthly_sales[monthly_sales['Order Year'] == year]
    ax.plot(year_data['Order Month'], year_data['Sales'],
            marker='o', linewidth=2.5, markersize=6,
            label=str(int(year)), color=colors[year])

ax.set_title('Monthly Sales Trend by Year', fontsize=16, fontweight='bold')
ax.set_xlabel('Month')
ax.set_ylabel('Total Sales ($)')
ax.set_xticks(range(1, 13))
ax.set_xticklabels(['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun',
                    'Jul', 'Aug', 'Sep', 'Oct', 'Nov', 'Dec'])
ax.legend(title='Year')
plt.tight_layout()
plt.savefig('chart3_monthly_trend.png', dpi=150, bbox_inches='tight')
plt.show()
print(" Chart 3 saved as 'chart3_monthly_trend.png'")

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CHART 3 - MONTHLY SALES TREND

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	Order Year	Order Month	Sales
0	2014	1	17712.1660
1	2014	2	5708.2320
2	2014	3	56783.2050
3	2014	4	30832.9450
4	2014	5	25138.0610
5	2014	6	26592.7436
6	2014	7	33229.1340
7	2014	8	34644.2855
8	2014	9	62787.2458
9	2014	10	26313.3340
10	2014	11	61550.3485
11	2014	12	62811.8660

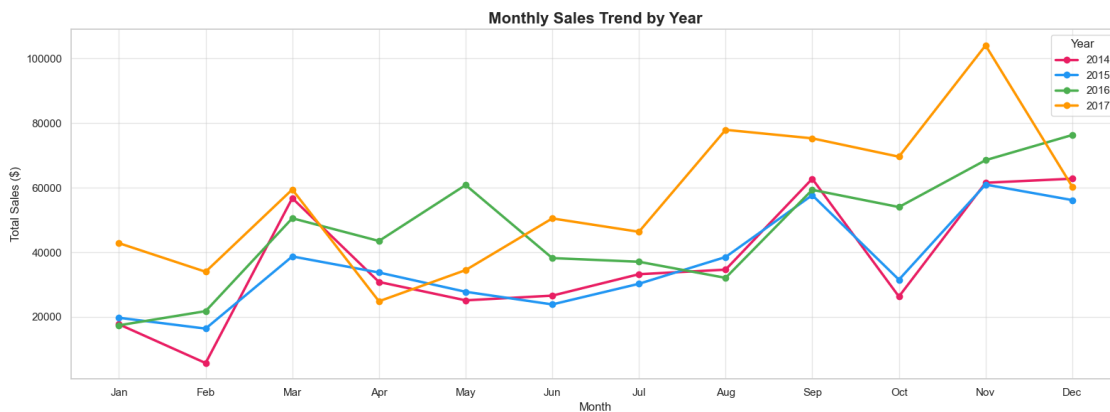


Chart 3 saved as 'chart3_monthly_trend.png'

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[9]: print("=" * 50)
print("CHART 4 - PROFIT BY SUB-CATEGORY")
print("=" * 50)

# Calculate total profit per sub-category and sort
subcategory_profit = df.groupby('Sub-Category')['Profit'].sum().reset_index()
subcategory_profit = subcategory_profit.sort_values('Profit', ascending=True)
print(subcategory_profit)

# Create colors - red for negative profit, green for positive
colors = ['#F44336' if x < 0 else '#4CAF50' for x in
         ↪subcategory_profit['Profit']]

# Create the chart
fig, ax = plt.subplots(figsize=(14, 8))

bars = ax.barh(subcategory_profit['Sub-Category'],
               subcategory_profit['Profit'], color=colors)

ax.set_title('Total Profit by Sub-Category', fontsize=16, fontweight='bold')
ax.set_xlabel('Total Profit ($)')
ax.set_ylabel('Sub-Category')
ax.axvline(x=0, color='black', linewidth=1.5)

# Add value labels
for bar, value in zip(bars, subcategory_profit['Profit']):
    ax.text(value + (500 if value >= 0 else -500),
            bar.get_y() + bar.get_height()/2,
            f'${value:,.0f}', va='center',
            ha='left' if value >= 0 else 'right',
            fontsize=9, fontweight='bold')

plt.tight_layout()
plt.savefig('chart4_subcategory_profit.png', dpi=150, bbox_inches='tight')
plt.show()
print(" Chart 4 saved as 'chart4_subcategory_profit.png'")
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CHART 4 - PROFIT BY SUB-CATEGORY
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	Sub-Category	Profit
16	Tables	-17417.5297
4	Bookcases	-3140.2376

15	Supplies	-1040.2771
8	Fasteners	990.7157
10	Labels	4944.0075
2	Art	6332.6718
7	Envelopes	6765.8879
11	Machines	9306.0862
9	Furnishings	12176.7773
1	Appliances	17523.7029
14	Storage	20682.7772
5	Chairs	23617.7082
3	Binders	31858.7292
12	Paper	32030.1987
0	Accessories	39998.7041
13	Phones	42833.2665
6	Copiers	46512.5796

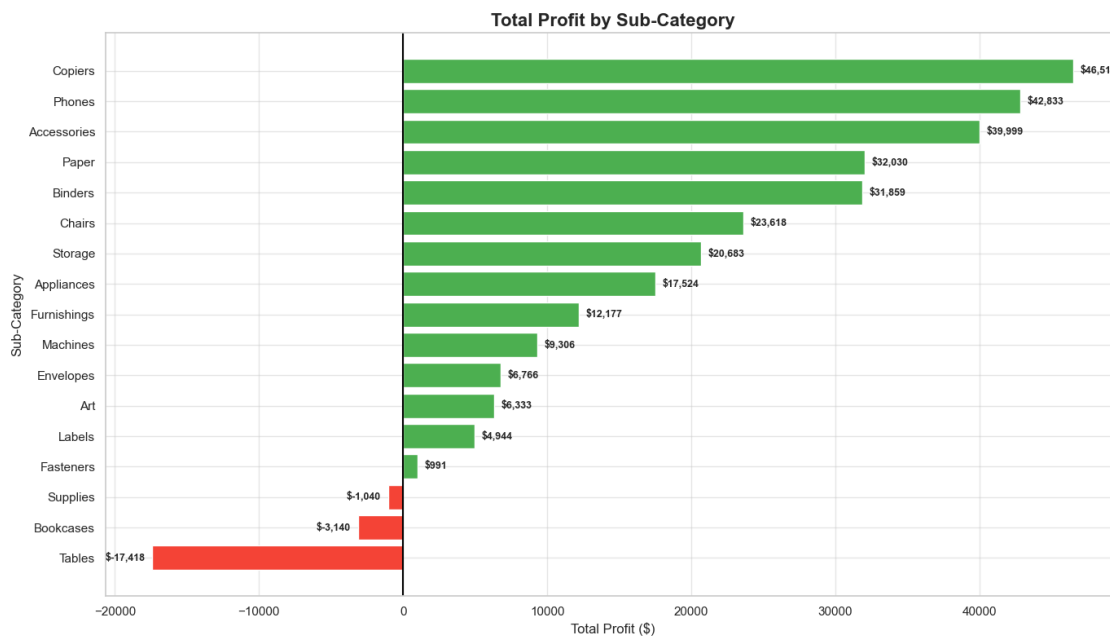


Chart 4 saved as 'chart4_subcategory_profit.png'

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[10]: print("=" * 50)
print("CHART 5 - CUSTOMER SEGMENT ANALYSIS")
print("=" * 50)

# Calculate sales and profit per segment
segment_data = df.groupby('Segment')[['Sales', 'Profit']].sum().reset_index()
print(segment_data)

# Create the chart
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fig, axes = plt.subplots(1, 3, figsize=(18, 6))

# Pie chart - Sales by Segment
axes[0].pie(segment_data['Sales'],
            labels=segment_data['Segment'],
            autopct='%1.1f%%',
            colors=['#2196F3', '#4CAF50', '#FF9800'],
            startangle=90,
            explode=(0.05, 0.05, 0.05))
axes[0].set_title('Sales by Segment', fontweight='bold')

# Pie chart - Profit by Segment
axes[1].pie(segment_data['Profit'],
            labels=segment_data['Segment'],
            autopct='%1.1f%%',
            colors=['#2196F3', '#4CAF50', '#FF9800'],
            startangle=90,
            explode=(0.05, 0.05, 0.05))
axes[1].set_title('Profit by Segment', fontweight='bold')

# Bar chart - Orders count by Segment
segment_orders = df.groupby('Segment')['Order ID'].count().reset_index()
axes[2].bar(segment_orders['Segment'], segment_orders['Order ID'],
            color=['#2196F3', '#4CAF50', '#FF9800'])
axes[2].set_title('Number of Orders by Segment', fontweight='bold')
axes[2].set_xlabel('Segment')
axes[2].set_ylabel('Number of Orders')
for i, v in enumerate(segment_orders['Order ID']):
    axes[2].text(i, v + 20, str(v), ha='center', fontweight='bold')

plt.suptitle('Customer Segment Analysis', fontsize=16, fontweight='bold')
plt.tight_layout()
plt.savefig('chart5_segment.png', dpi=150, bbox_inches='tight')
plt.show()
print(" Chart 5 saved as 'chart5_segment.png'")

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CHART 5 - CUSTOMER SEGMENT ANALYSIS

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	Segment	Sales	Profit
0	Consumer	1.067228e+06	126473.6054
1	Corporate	6.541229e+05	89693.5566
2	Home Office	3.979223e+05	57808.6064

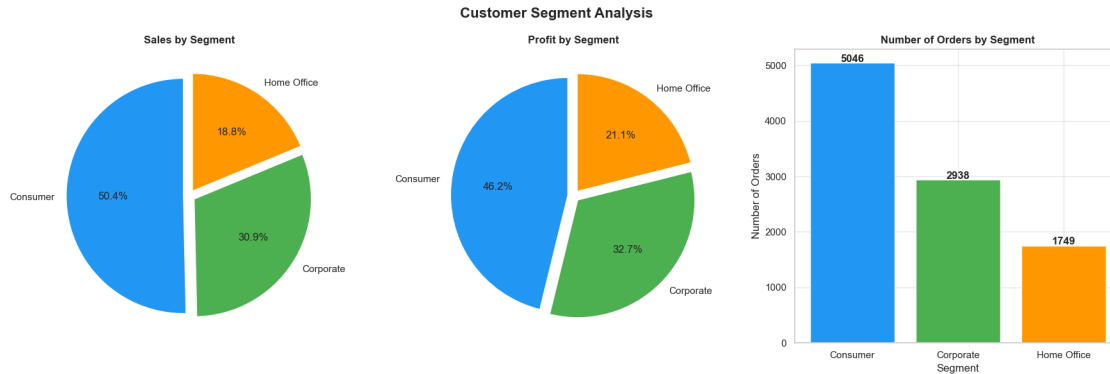


Chart 5 saved as 'chart5_segment.png'

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[11]: print("=" * 50)
print("CHART 6 - DISCOUNT VS PROFIT")
print("=" * 50)

# Create the chart
fig, axes = plt.subplots(1, 2, figsize=(16, 6))

# Scatter plot - Discount vs Profit
axes[0].scatter(df['Discount'], df['Profit'],
                alpha=0.3, color='#2196F3', s=10)
axes[0].axhline(y=0, color='red', linewidth=1.5, linestyle='--')
axes[0].axvline(x=0, color='black', linewidth=1)
axes[0].set_title('Discount vs Profit (Scatter)', fontweight='bold')
axes[0].set_xlabel('Discount')
axes[0].set_ylabel('Profit ($)')

# Average profit by discount level
discount_profit = df.groupby('Discount')['Profit'].mean().reset_index()
axes[1].bar(discount_profit['Discount'].astype(str),
            discount_profit['Profit'],
            color=['#F44336' if x < 0 else '#4CAF50'
                  for x in discount_profit['Profit']])
axes[1].axhline(y=0, color='black', linewidth=1.5)
axes[1].set_title('Average Profit by Discount Level', fontweight='bold')
axes[1].set_xlabel('Discount Level')
axes[1].set_ylabel('Average Profit ($)')
axes[1].tick_params(axis='x', rotation=45)

plt.suptitle('Impact of Discount on Profit', fontsize=16, fontweight='bold')
plt.tight_layout()
plt.savefig('chart6_discount_profit.png', dpi=150, bbox_inches='tight')
plt.show()
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print(" Chart 6 saved as 'chart6_discount_profit.png'")
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CHART 6 - DISCOUNT VS PROFIT

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Chart 6 saved as 'chart6_discount_profit.png'

```
[12]: print("=" * 50)
print("CHART 7 - YEARLY SALES GROWTH")
print("=" * 50)

# Calculate total sales and profit per year
yearly_data = df.groupby('Order Year')[['Sales', 'Profit']].sum().reset_index()

# Calculate growth percentage
yearly_data['Sales Growth %'] = yearly_data['Sales'].pct_change() * 100
yearly_data['Profit Growth %'] = yearly_data['Profit'].pct_change() * 100
print(yearly_data)

# Create the chart
fig, axes = plt.subplots(1, 2, figsize=(16, 6))

# Bar chart - Yearly Sales & Profit
x = range(len(yearly_data))
width = 0.35
bars1 = axes[0].bar([i - width/2 for i in x], yearly_data['Sales'],
                    width, label='Sales', color='#2196F3')
bars2 = axes[0].bar([i + width/2 for i in x], yearly_data['Profit'],
                    width, label='Profit', color='#4CAF50')

axes[0].set_title('Yearly Sales & Profit', fontweight='bold')
axes[0].set_xlabel('Year')
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axes[0].set_ylabel('Amount ($)')
axes[0].set_xticks(x)
axes[0].set_xticklabels(yearly_data['Order Year'].astype(int))
axes[0].legend()

for bar in bars1:
    axes[0].text(bar.get_x() + bar.get_width()/2, bar.get_height() + 1000,
                 f'${bar.get_height():,.0f}', ha='center', fontsize=8,
                 fontweight='bold')
for bar in bars2:
    axes[0].text(bar.get_x() + bar.get_width()/2, bar.get_height() + 1000,
                 f'${bar.get_height():,.0f}', ha='center', fontsize=8,
                 fontweight='bold')

# Line chart - Growth %
axes[1].plot(yearly_data['Order Year'].astype(int),
             yearly_data['Sales Growth %'],
             marker='o', linewidth=2.5, markersize=8,
             color='#2196F3', label='Sales Growth %')
axes[1].plot(yearly_data['Order Year'].astype(int),
             yearly_data['Profit Growth %'],
             marker='s', linewidth=2.5, markersize=8,
             color='#4CAF50', label='Profit Growth %')
axes[1].axhline(y=0, color='red', linewidth=1.5, linestyle='--')
axes[1].set_title('Yearly Growth Rate (%)', fontweight='bold')
axes[1].set_xlabel('Year')
axes[1].set_ylabel('Growth %')
axes[1].legend()

for i, row in yearly_data.iterrows():
    if pd.notna(row['Sales Growth %']):
        axes[1].text(int(row['Order Year']), row['Sales Growth %'] + 1,
                     f"{row['Sales Growth %']:.1f}%", ha='center', fontsize=9)
    if pd.notna(row['Profit Growth %']):
        axes[1].text(int(row['Order Year']), row['Profit Growth %'] - 3,
                     f"{row['Profit Growth %']:.1f}%", ha='center', fontsize=9)

plt.suptitle('Yearly Sales & Profit Growth', fontsize=16, fontweight='bold')
plt.tight_layout()
plt.savefig('chart7_yearly_growth.png', dpi=150, bbox_inches='tight')
plt.show()
print(" Chart 7 saved as 'chart7_yearly_growth.png'")

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CHART 7 - YEARLY SALES GROWTH

Order Year	Sales	Profit	Sales Growth %	Profit Growth %
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0	2014	444103.5664	45722.8614	NaN	NaN
1	2015	435508.4789	59143.4132	-1.935379	29.351951
2	2016	559813.8082	84074.4893	28.542574	42.153597
3	2017	679847.3222	85035.0045	21.441685	1.142457

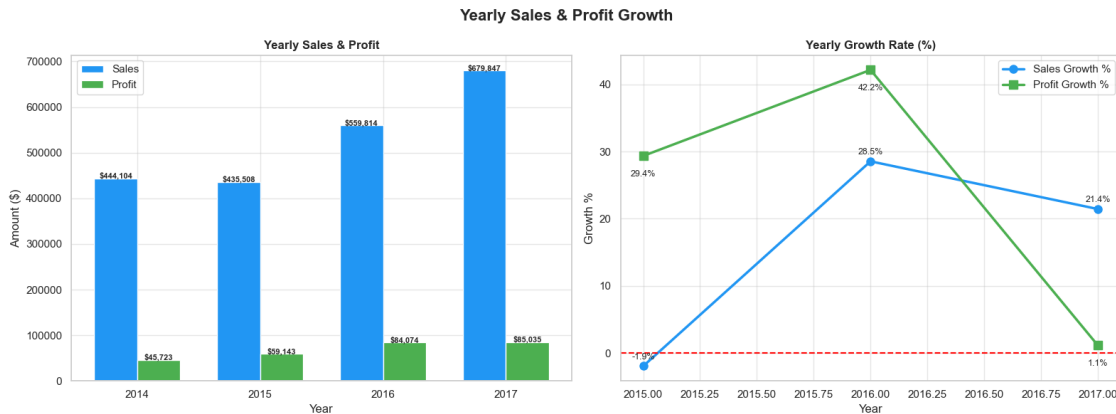


Chart 7 saved as 'chart7_yearly_growth.png'

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[13]: print("=" * 50)
print("CHART 8 - CORRELATION HEATMAP")
print("=" * 50)

# Select numeric columns for correlation
numeric_cols = ['Sales', 'Profit', 'Quantity', 'Discount']
correlation = df[numeric_cols].corr()
print("Correlation Matrix:")
print(correlation)

# Create the heatmap
fig, ax = plt.subplots(figsize=(8, 6))

sns.heatmap(correlation,
            annot=True,
            fmt='.2f',
            cmap='RdYlGn',
            center=0,
            vmin=-1, vmax=1,
            square=True,
            linewidths=1,
            ax=ax)

ax.set_title('Correlation Heatmap\n(Sales, Profit, Quantity, Discount)',
            fontsize=14, fontweight='bold')
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plt.tight_layout()
plt.savefig('chart8_correlation.png', dpi=150, bbox_inches='tight')
plt.show()
print(" Chart 8 saved as 'chart8_correlation.png'")
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===== CHART 8 - CORRELATION HEATMAP =====

Correlation Matrix:

	Sales	Profit	Quantity	Discount
Sales	1.000000	0.445305	0.178465	-0.027021
Profit	0.445305	1.000000	0.067518	-0.216543
Quantity	0.178465	0.067518	1.000000	0.004040
Discount	-0.027021	-0.216543	0.004040	1.000000

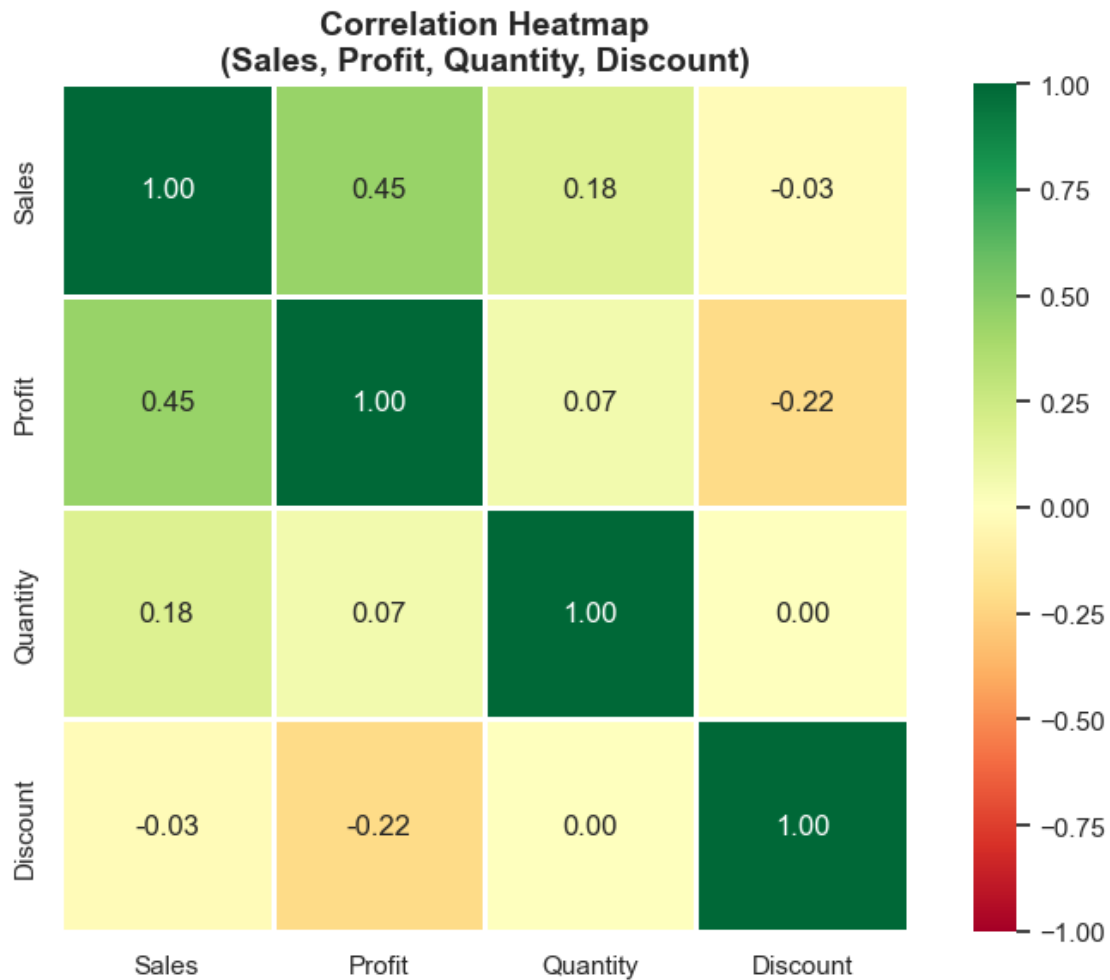


Chart 8 saved as 'chart8_correlation.png'